

29d SGD - Random Variable Convergence Worksheet

Wednesday, April 17, 2024

11:17 AM

Let $\Omega = [0, 1]$ w/ $\mathbb{P}$ the uniform measure on Ω , i.e. $\mathbb{P}([a, b]) = \mathbb{P}(a, b) = b - a$
 $\forall 0 \leq a \leq b \leq 1$

$$\text{Let } e_n(\omega) := \begin{cases} n^\alpha & 0 \leq \omega < 1/n \\ 0 & 1/n \leq \omega \leq 1 \end{cases}$$

$$\text{So } (*) \ e_n = \begin{cases} n^\alpha & \text{w.p. } 1/n \\ 0 & \text{w.p. } 1 - 1/n \end{cases}$$

Activity: fill in the table (answers in red)

α	L^p (which p ?)	almost sure	in probability
0	$\forall p \in [1, \infty)$ not $p = \infty$	yes	yes
$\frac{1}{2}$	$\forall p \in [1, 2)$	yes	yes
1	No p	yes	yes

justification: either direct proof or follows from "a.s."

Q: Can you define f_n that also satisfy (*) but don't converge in the same way?

$$\text{A: let } f_n = e_n \text{ if } n \text{ even, } f_n(\omega) = \begin{cases} n^\alpha & 1 - 1/n < \omega \leq 1 \\ 0 & 0 \leq \omega \leq 1 - 1/n \end{cases} \text{ if } n \text{ odd}$$

Then some L^p convergence but doesn't converge a.s. for any α (so for $\alpha = 0, 1/2$ it still converges in probability. For $\alpha = 1$ it also converges in prob. via direct calculation)